

Passive Opportunity Scanner

Overview

The **Passive Opportunity Scanner** is an AI-powered web application that helps job seekers discover relevant job opportunities based on their resume. Instead of manually searching across multiple job platforms, users simply upload their resume, and the system automatically scans various sources, analyzes job relevance, and presents the most suitable opportunities.

Project Workflow

1. Resume Upload

Users begin by uploading their resume in either **PDF** or **DOCX** format. The application extracts the content from the uploaded file using a PDF parser and document processing techniques.

2. Skill Extraction

After parsing the resume, the system identifies and extracts the candidate's key skills, technologies, and qualifications. These extracted skills are then used as the foundation for searching and evaluating job opportunities.

3. Job Scanning

Once the user clicks the **"Scan Jobs"** button, the backend initiates a multi-source job search using **LangGraph**. Job listings are collected from several platforms, including:

- Tavily
- SerpAPI
- Google Search
- LinkedIn

This allows the application to gather a wide range of relevant job postings from different sources.

4. Job Matching

The collected job descriptions are analyzed against the user's extracted skills. Each job is assigned a matching score based on its relevance to the candidate's profile. Only jobs with a **matching score of 70% or higher** are displayed, ensuring that users receive highly relevant opportunities.

5. Job Summary

For each matched opportunity, the application provides a concise summary, enabling users to quickly understand the role, required skills, and overall suitability without reading lengthy job descriptions.

6. AI-Assisted Job Application

The application includes an AI-powered feature that helps users generate a professional job application or cover letter tailored to the selected job. Multiple large language models, including **OpenAI, Gemini, Claude, and Grok**, are integrated to provide intelligent and context-aware content generation.

7. View Original Job Description

Users can access the complete job description directly from the original source platform through the provided job link, allowing them to review all details before applying.

8. Email Summary

The system also includes an email summary feature. After completing the job scan, users can send a summary of the matched opportunities to any email address through the dashboard. The application connects to an email server, sends the summary automatically, and the user can verify the received email in their inbox.

Technologies Used

Backend

- FastAPI
- LangGraph
- PostgreSQL with PGVector

Frontend

- React

AI Models

- OpenAI
- Gemini
- Claude
- Grok

External Services

- Tavily
- SerpAPI
- Google Search
- LinkedIn

Key Features

- Resume upload in PDF and DOCX formats.
- Automatic skill extraction from resumes.
- Multi-platform job scanning.
- AI-powered job relevance analysis.
- Display of jobs with a matching score above 70%.
- AI-generated job applications and cover letters.
- Direct access to the original job description.
- Email summary of matched job opportunities.
- Interactive dashboard for managing job search results.